THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET B

Chapter 10: **Circles** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat figures wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The maximum number of parallel tangents a circle can have in a given direction is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** infinitely many

2. From a point 10 cm away from the centre, the length of the tangent to a circle of radius 6 cm is: 1

(a) 8 cm **(b)** 6 cm **(c)** 4 cm **(d)** 16 cm

3. The common point of a tangent and a circle is called the: 1

(a) centre **(b)** point of contact **(c)** end point **(d)** chord

4. If tangents at P and Q meet at T and $\angle POQ = 140^\circ$, then $\angle PTQ$ is: 1

(a) 40° **(b)** 50° **(c)** 70° **(d)** 140°

5. A student states: “A tangent to a circle can pass through its centre.” The correct response is: 1

(a) Correct

(b) Incorrect — a line through the centre meets the circle at two points

(c) Correct for small circles

(d) Cannot be determined

6. PA and PB are tangents from P to a circle with centre O and $\angle APB = 50^\circ$. Then $\angle OAB$ is: 1

(a) 25° **(b)** 50° **(c)** 65° **(d)** 40°

7. A quadrilateral ABCD circumscribes a circle with $AB = 8$, $BC = 9$, $CD = 7$ cm. Student A says $AD = 6$ cm; Student B says $AD = 8$ cm. Then: 1

(a) Student A is correct **(b)** Student B is correct

(c) Both are correct **(d)** Both are wrong

8. The length of the tangent from a point A 5 cm from the centre is 4 cm. The radius is: 1

(a) 3 cm **(b)** 4 cm **(c)** 5 cm **(d)** $\sqrt{41}$ cm

9. The distance between two parallel tangents of a circle of radius 4 cm is: 1

(a) 4 cm **(b)** 8 cm **(c)** 16 cm **(d)** 2 cm

10. Two tangents inclined at 90° are drawn to a circle of radius 3 cm. The length of each tangent is: 1

(a) 3 cm **(b)** $3\sqrt{2}$ cm **(c)** 6 cm **(d)** 1.5 cm

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. PQ and PR are tangents to a circle with centre O such that $\angle QPR = 46^\circ$. Find $\angle QOR$. 2

12. Two concentric circles have radii 10 cm and 6 cm. Find the length of the chord of the larger circle which touches the smaller circle. 2

13. Prove that the perpendicular at the point of contact to the tangent of a circle passes through the centre. 2

14. PA and PB are tangents to a circle of radius 5 cm with $\angle APB = 90^\circ$. A student says $PA = 5\sqrt{2}$ cm. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Prove that the tangent at any point of a circle is perpendicular to the radius through the point of contact. 3

16. XY and X′Y′ are two parallel tangents to a circle with centre O, and another tangent AB with point of contact C intersects XY at A and X′Y′ at B. Prove that $\angle AOB = 90^\circ$. 3

17. Prove that the opposite sides of a quadrilateral circumscribing a circle subtend supplementary angles at the centre of the circle. 3

OR

Prove that the chord of the larger of two concentric circles which touches the smaller circle is bisected at the point of contact. 3

18. Prove that the tangents drawn at the ends of a chord of a circle make equal angles with the chord. 3

OR

From an external point P, tangents PA and PB are drawn to a circle with centre O. If $PA = 12$ cm and $\angle APB = 60^\circ$, find the length of chord AB and $\angle OAB$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Prove that the lengths of tangents drawn from an external point to a circle are equal. Using this, if a circle touches side BC of $\triangle ABC$ at P and AB, AC produced at Q and R respectively, show that $AQ = \frac{1}{2}(\text{perimeter of } \triangle ABC)$. 5

OR

A circle is inscribed in $\triangle ABC$ with $AB = 12$ cm, $BC = 8$ cm and $AC = 10$ cm, touching AB, BC, CA at D, E, F. Find AD, BE and CF. Verify that $AD + BE + CF$ equals the semi-perimeter. 5

20. **Case study:** A designer makes a kite-shaped frame ABCD around a circular badge so that all four sides touch it. $AB = AD = 10$ cm and $BC = CD = 15$ cm. 5

(i) Check whether $AB + CD = AD + BC$.

(ii) If the badge touches AB at P with $AP = 4$ cm, find the tangent lengths from B, C and D.

(iii) Prove that for any quadrilateral circumscribing a circle, $AB + CD = AD + BC$.

OR

Two tangents TP and TQ are drawn from an external point T to a circle with centre O. 5

(i) Prove that OT is the perpendicular bisector of PQ.

(ii) If $OP = 5$ cm and $OT = 13$ cm, find the length of PQ.

END OF QUESTION PAPER — **Total: 40 Marks**